Sonify Abstract

The user is provided with a virtual 2-D space that is to be manipulated to craft a dynamic soundscape. The goal of developing this project is to provide a tool to a general audience with which they can develop a musical intuition, based on already strong standards in visualization and touch interactions. Rather than relying on musical standards such as rhythm and key, the app presents users with a space to experiment and discover musical concepts on their own.

In the app, the user is presented with a blank canvas, which should be thought of as a virtual space, like looking down into a room from the ceiling. There are 3 types objects that can be placed in this space: sound *Emitters*, sound *Receptors*, and *Walls*. *Emitters* are the source of all sounds, either generated by a function or played back from a recording. There can be several *Emitters* present at once, and users can manually modify their properties through a set of controls. *Receptors* directionally receive the sound given off by *Emitters* and are the final stop for the sound. Finally, *Walls* are paths that users draw freely in the space, and they modify sound as it passes through or bounces off of them. There are different "materials" that *Walls* can be made of that will apply different effects to the sound they encounter. The proximities and qualities of all these objects determine the final output, and because any number of these objects can exist anywhere on the canvas, there is a huge number of possibilities for what the user can produce. As users draw and change parameters, the sound changes in real time, allowing constant feedback from their interactions.

The expressive nature of the interface also allows it to be used in a performance setting. Once users have familiarized themselves with the techniques to operate it, they can conduct a live performance by drawing *Walls* or by moving, activating and deactivating *Emitters* and *Receptors* in real time. Because interacting with the app creates a visual interpretation of the sound, the interface could also be projected to provide visual feedback for an audience.